

- CHAPTER 11 -

PREHABILITATION, ISOLATION, FLEXIBILITY, AND COOL DOWN

Now that you have learned what composes the core of a workout, we can talk about what comes at the end: prehabilitation work, isolation work, flexibility work, and cool down. It is important to have goals within this category, especially for prehabilitation and flexibility. You should perform these at the end of every workout to help your body relax and prevent overuse injuries.

It is vital to keep your joints, tendons, and muscles moving well. Good quality body tissues should not hurt or feel sore or painful when you apply pressure or massage them. If you place your hands on relaxed muscles or tendons they should be pliable and easy to move. Unfortunately, it is common for them to feel tight, crampy, and bumpy.

Your joints should feel good when you use them. Your muscles should be soft and pliable. Take a moment to close your eyes and move your body. You should feel good and unrestricted in movement. Think of an athlete like Michael Phelps. Before he swims in a competition, he performs many arm circles and throws his arms back and forth to loosen them up. You can see that his muscles are soft and pliable; they move unhindered. That is your goal.

The sample routine example of prehabilitation, Isolation, Flexibility, and Cool Down

- 3x1 minute sets of Rice Bucket for your wrists
- 3x10 Biceps Curls
- 3-5x30s Split Holds
- 3-5x30s German Hangs
- 3-5x20s Back Bridges
- 1 minute of Deep Breathing (in through the nose, out through the mouth)

PREHABILITATION WORK

Prehabilitation work refers to the part of your workout that is focused on injury prevention. Prehabilitation exercises are focused on correcting imbalances and preventing injuries from occurring by performing specific exercises on your weak links. *Rehabilitation work* can be done in either conjunction with prehabilitation work

or in separate sessions. If you are not using the injured body part in your workout, it is a good idea to add your rehabilitation and prehabilitation exercises to the end of your routine. Alternatively, these exercises can be used to warm up your tissues before working out. Consider utilizing prehabilitation if you have had previous injuries to certain joints or if you suspect an injury may be developing due to twinges of pain and/or prolonged soreness of muscles, tendons, or joints.

The best way you can think of prehabilitation is that it is an extension of rehabilitation. Prehabilitation is derived from “pre-rehabilitation,” or performing rehabilitative work before you acquire an injury. It is an extension of rehabilitation; both in the context of pre-injury and post-injury rehabilitation state: *There is an awareness of discomfort in certain muscles, connective tissues, or joints that may potentially develop into an issue, so you are heading it off with specific corrective exercises.*

You will pass through a similar stage whether you acquire an injury or not, so it is important to learn to identify the nature of pre-injury vs. post-injury state. It could be a result of imbalance, weakness, overuse, or other factors. After you identify the situation, you need to be proactive in correcting imbalances, strengthening, or resting to bring your tissues, tendons, and joints back into a healthy state.

Prehabilitation is always used in two distinct circumstances. The first is as an extension of rehabilitation where you previously had an injury, such as tendonitis. In this example, the tendonitis is currently rehabilitated to the point where you do not feel pain, but excessive volume or intensity in your workout could re-injure it. In other words, this part of your body is still vulnerable to injury. You may need to program around this area in your general training. The second circumstance where you would use prehabilitation is when you are in a state of pre-injury. It is better to catch an impending injury before it devolves into pain, muscle atrophy, and other negative consequences. In this instance, prehabilitation can help restore your tissues to a normal, healthy state. If your tissues are overworked or “under-recovered,” you may feel discomfort, achiness, soreness, tightness, or twinges of pain. You will want to dial down your training, analyze what is taking place, and take the necessary steps to correct it.

The shoulder is a common joint where prehabilitation work may be needed. If your shoulders need additional stabilization exercise, perform exercises for your rotator cuffs or stability exercises such as Turkish get-ups for the glenohumeral joint. Additionally, if there is tightness around your scapulas, you may want to perform specific scapular stabilization exercises like scapular retractions, depressions, elevations, and protractions. Thoracic mobilization may be necessary as well. All of the different parts of your shoulder may play a role in your need for prehabilitation work.

The main reason to place specific prehabilitation work at the *end* of the workout is that performing these exercises before a workout will fatigue you. If you desire to gain strength with an exercise you need to fatigue your primary muscles during a difficult movement. When your stabilizer muscles are fatigued, they will fail much more rapidly during compound exercises, which limits the workload of your primary movers and may increase your chance of muscle strains and other injuries.

One example is biceps curls and pull-ups. If you fatigue your biceps before performing your pull-ups, it is likely that your pull-up numbers will suffer. Performance decreases when you perform prehabilitation work on a certain area of the body prior to trying to use that part to perform compound exercises for strength.

ISOLATION WORK

Isolation work is used for many purposes. It can be used as prehabilitation work in order to strengthen connective tissues, for strength work, for aesthetic hypertrophy, or a combination of both prehabilitation and strength work, where you are simultaneously strengthening your muscles and preventing injury by correcting imbalances and recognizing weak links in your body.

A 3x10 set and repetition scheme is typically used for hypertrophy of the muscle because of the repetition range. In the example routine, we used a set of 3x10 biceps curls. This 3x10 set may also be used for aesthetics—in other words, you might do a bit more isolation work to get bigger biceps. However, if you have been performing pull-ups for a long time and your biceps tend to be your weak link, the 3x10 biceps curls may also be used to bring up biceps strength in order to perform better on pull-ups. In this case, the hypertrophy in the biceps will assist in strength because we know that *strength = neural adaptations * muscle cross sectional area*. In the case of strength without much of an increase in muscle mass, the biceps curls may be adjusted to 5x5 or another lower repetition range with more sets, to bias toward strength.

You can also use an isolation exercise (such as biceps curls) for prehabilitation. Specific strengthening work (such as the back lever or any other rings work) can often put a lot of strain on the biceps tendon, leading to soreness or potentially tendonitis in the biceps tendon. If that is the case, you would use an exercise that trains the biceps tendon in a much higher repetition range to strengthen connective tissues.

The goal is to not increase intensity, as the tissues are already slightly compromised and higher intensity would only make the injury worse. To illustrate: if you have a hamstring strain, performing a high-intensity exercise, like sprinting, is a bad idea. Rather, you should be training not-to-failure in order to avoid potential injury complications. Connective tissues such as tendons, ligaments, and cartilage tend to respond better to higher repetitions because of the increase in blood flow that assists with healing. The mechanical tension is still high enough to stimulate repair and remodeling without the potential forces to further injure it. Utilize not-to-failure training, very light weight, very high repetitions, and a lower number of sets in order to strengthen your connective tissues, for example, do biceps curls with a very low weight (such as five pounds) for three sets of 30-50 repetitions, not to failure.

As you can see, prehabilitation work and isolation work have some overlap. Isolation can be used with prehabilitation to bring up certain weak links or strengthen connective tissues by modifying intensity and volume to match.

COMBINING PREHABILITATION AND ISOLATION WORK

Prehabilitation and isolation work are easy to use in combination. If your wrists are becoming sore from performing too many handstands, regain mobility by using isolation work for your wrists in order to help them withstand the stress of handstands. You may use wrist curls, moving the wrist around in a rice bucket, wrist pushups, or other types of exercises to shore up your wrists against injury. What you *do not* want to do is continue performing handstands through the soreness and pain that follows. This will eventually lead to injury.

Similarly, if you have soreness or pain in your shoulders, you may need some specific prehabilitation-focused isolation work for the scapula and shoulder joint. Earlier, we gave an example of specific rotator cuff work that could be followed up with specific scapular isolation strengthening.

For extremity joints such as the wrists, ankles, elbows, and knees, it is likely that wherever your discomfort or soreness is present, that is where the prehabilitation and isolation work need to be focused. On the other hand, the joints closer to the spine tend to interact with the torso in a way that may confound easy identification. The shoulder is an obvious case. For instance, limitation in scapular mobility may manifest as shoulder pain. The limitation of the scapula to move may cause the shoulder to take up extra effort by compensation or cause specific parts of the shoulder to rub against one another, causing irritation and discomfort. In this case, increasing scapular mobility may be a better solution over rotator cuff work or a combination of the two.

Specific work to the area in question should begin to help within three to five workouts or two weeks. This will give you enough time to remove the aggravating exercises from your routine and begin prehabilitation and isolation work to address the issue at hand. If the issue does not improve, there is likely something else going on that is interfering with improvements, or the fundamental issue itself is not being addressed.

An example of this is a tight back and foam rolling. An athlete may notice they have a tight back and continuously use a foam roller or other massage-like implements to remove the knots and pain. Their back will improve for a period of time, but eventually stop improving. At this point continuing foam rolling will not help; the tightness is actually being caused by something else. The three most common causes of tightness in the back are pain, instability, or weakness. You may have one of these things happening, two of them happening, or all of them happening at once. Pain and forceful movement to the end of your range of motion from spraining an ankle causes the muscles around the ankle to become tight as the body responds to the injury. A person that is double jointed and hypermobile (such as contortionists) will have tight muscles because their body responds to the instability of hypermobile joints by tightening the muscles to prevent injury. If you have a weak back, your muscles may become tight to protect from injury.

Now, in the case of foam rolling, you still have a tight back. This tightness could be caused by lingering pain from a previous injury. People with pain usually have significant tightness, which “magically” clears up as they heal from the injury. Instability could be another reason why your back is tight. Specific stability work for the spine may be the solution, rather than more foam rolling. Alternatively, your back could be weak and your muscles are tightening to protect it from injury. In that case, perform exercises to strengthen your back.

High-repetition kettlebell swings are known to help significantly with back tightness and pain. Why? Because kettlebell swings with light weights force your core muscles to stabilize your spine while simultaneously providing a stimulus for the back to become stronger under load. This corrects both instability and weaknesses at the same time. Over time, the pain will clear up, leaving one with a stronger, pain-free back. Likewise, high repetition reverse hyperextensions tend to be a good exercise as well due to the same benefits. Obviously, you should not attempt to solve your back pain with kettlebell swings or reverse hyperextensions without consulting a medical professional. This should help you understand the concept.

In general, if your injury is getting worse and nothing you are doing to fix it is working, should stop exercising and consult a medical professional immediately!

Here are some general recommendations for prehabilitation and isolation work:

- Isolation work can be performed after a workout for hypertrophy if you are interested in aesthetics. This should utilize standard strength and hypertrophy sets and rest times.
- Prehabilitation, usually in the form of isolation work, is meant to strengthen weaknesses, fix instability, or correct other issues for a particular area. This should use low intensity and high repetitions to prevent re-injury and retrain the particular areas.
- Prehabilitation for your extremities (such as the wrists, elbows, ankles, and knees) tends to be straightforward. More factors may be involved when dealing with the spine, hips, and shoulders. Use moderate to high repetitions.
- Exercise is a potentially dangerous activity. Although this book gives you tools and concepts, you may not be able to use them correctly. If you have any questions or concerns, you should consult the appropriate medical professional before beginning an exercise routine. This could be an orthopedic doctor, sports medicine doctor, and/or physical therapist.

As far as programming for prehabilitation goes, generally aim for:

- Longer rests between sets; 2-3 minutes or more
- Slower tempos for prehabilitation or isolation work, such as 5121
- Higher repetitions, such as 15-40
- Not to failure work

The goal is to not push your connective tissues too much by going to failure, moving quickly, or decreasing rest times. Instead, you should use an exercise to “injury proof” the area. The high-repetition sets mentioned earlier are recommended, but there are exceptions. Higher repetitions for prehabilitation work tend to work well, especially for potential tendonitis or strain issues. However, if you know the issue is a specific weakness you will want to use lower repetitions (such as in the 5-8 range) to increase strength.

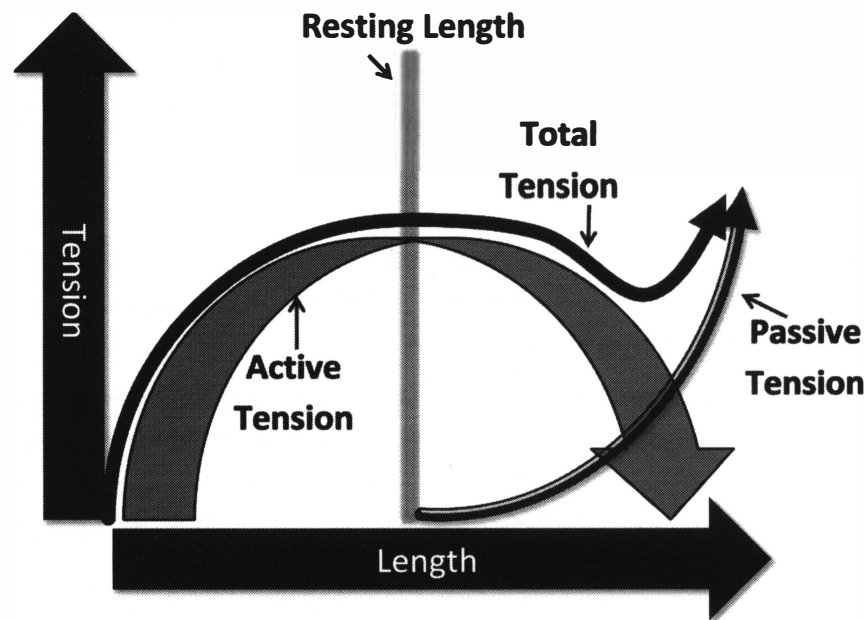
When referring to upper-body training there are five main areas to consider: the hips, spine, shoulders, elbows, and wrists. These joints and their surrounding musculature are the main areas in bodyweight exercises that you need to safeguard. Many specific exercises are located later in this book. There are many others. If you have your own set that you like, by all means, use them.

FLEXIBILITY WORK AND COOL DOWN

Flexibility work and cool down make up the final part of a routine. Generally speaking, flexibility is ideal for the cool down, but the cool down can have additional light activity like jogging, jumping rope, or mobility work. If you have anything that does not fit in your routine because it is oddball, you can place it in this section.

Now let’s learn why flexibility exercises work. When you understand how something works on a conceptual level, you can make tweaks to your routine in order to progress.

In the first chapter of this book, we talked about muscle spindles. They are the structures within your muscles that regulate muscle length. If you lengthen a muscle to the limit of your range of motion, your muscle spindles will send feedback to your brain to contract the muscles in order to provide tension. This generates passive tension within the muscles, as seen in the picture below.



The nervous system has *afferent* and *efferent* fibers. *Afferent* fibers are sensor fibers that provide feedback to the nervous system via touch, pressure, or other changes within the body. *Efferent* fibers provide controlling feedback after the brain and/or nervous system has processed it. In conjunction with this are alpha, beta, and gamma afferent and efferent fibers. The Greek-letter-names of the fibers simply tell you their speed of transmission. In the case with muscle spindles there is gamma afferent feedback to the nervous system and brain when a muscle lengthens. After the afferent feedback has been processed there is alpha efferent and gamma efferent control to the muscles. The alpha efferent fiber control increases tension in the muscles and the gamma efferent control ensures that the muscle spindle reacts appropriately to the change in length.

- Gamma afferents → brain → alpha efferent + gamma efferent co-activation = alpha efferent contracting muscle + gamma efferent modulating muscle spindle length.

What occurs in flexibility and plyometric training is fusimotor system activation and modulation. The fusimotor system is composed of gamma and beta afferent and efferent fibers. The same process occurs above with the gamma afferents and processing, but the efferent feedback is slightly different. Stretching sends the gamma afferent feedback. The nervous system via the spinal cord and brain processes it. Alpha, beta, and gamma efferents are then sent back to the muscles.

- Flexibility training or plyometric training → gamma afferents → nervous system via spinal cord or brain → alpha, beta, and gamma efferents → alpha efferent contraction of the muscles + gamma efferent modulating muscle spindle length + beta efferents modulating muscle spindle sensitivity

Flexibility training aims to induce decreased muscle spindle sensitivity. In other words: when you train and your muscles start to lengthen and move to end range, your muscle spindles decrease sensitivity so that they respond later than in a previous stretch. You move further in your range of motion before the passive tension increases. This leads to gains in range of motion.

There is a division of static and dynamic control for beta efferent feedback to muscle spindles.

- Static = Muscle spindle sensitivity in static positions = muscle stretch at end range
- Dynamic = Muscle spindle sensitivity in dynamic movement = stretch shorten cycle = effectiveness in plyometrics

Flexibility training aims to decrease muscle spindle sensitivity to end range in static or slow movements to allow for increases in range of motion, which pushes back the range of passive tension. Plyometric training is targeted at increasing muscle spindle sensitivity in dynamic movements—those used in running and jumping.

It is vital to remember that pain inhibits decreasing muscle spindle sensitivity in flexibility training, and pain also inhibits increasing muscle spindle sensitivity in dynamic or plyometric training. This means that performing flexibility training into the pain range is extremely counterproductive.

To make this clear, let's take the example of a sprained ankle. A sprained ankle brings a joint to the edge of its range of motion very rapidly and often past it. The pain is typically at the joint itself and that is most athletes' biggest concern. However, think back to your last mild sprain. Walking on this mild or moderately sprained ankle is tough at first because it is stiff. This is your body providing feedback from nociceptors (pain receptors), as well as the muscle spindles. The feedback causes your body to stiffen up the muscles surrounding the area to protect it. In these cases, the muscle spindles undergo an immediate sensitization that makes the muscles stiff and resistant to move toward your range of motion. A sprained ankle will often have a severely limited range of motion, due to both pain and muscle stiffness.

A week or two after the injury, if there is no rehabilitation you will likely feel a limitation in moving toward the range of motion, even though the pain may be gone. It is the remnant of sensitization of the muscle spindles that is prohibiting you from moving toward your previous range of motion. If you are going to physical therapy, you are probably performing stretches and strengthening exercises to increase your range of motion and bring your strength level back to normal.

What often takes place during a session of painful stretching is a lower level of muscle spindle sensitization. The reason why general flexibility training does not work for some people is that they are performing it wrong. They are overstretching their muscles to the point of pain, which inhibits their process in the next session. They may be able to increase their range of motion in this first session; however, in the next session, they find during warm-ups that they cannot stretch any further than their previous session. This is because pain inhibited their muscle spindles from desensitizing, leaving them with the same range of motion they had in the previous session with no permanent gains.

Now that you know how the process works, let's go through some of the different methods of flexibility training.

FLEXIBILITY THROUGH STATIC STRETCHING

Standard static stretching works for most people in terms of flexibility, and it is most often performed in a normalized number of sets, with specific time limits.

- 3-6 sets of 30-60s holds for each exercise

Let's use the splits as an example. Move your left leg forward, right leg forward, and middle split each for 3-6 sets of 30-60 seconds. For each stretch, the method that works best is to move into the slight discomfort range and to focus on relaxing. As we discussed, moving into a range where you are experiencing pain is not beneficial. The pain signals to your nervous system to contract the muscles harder, and it does not allow your body to reduce its sensitivity to muscle lengthening. In the range where you experience a stretch and slight discomfort, hold for 30-60 seconds while utilizing deep breathing techniques. Allow your muscles to lengthen by relaxing as you exhale.

This type of generalized stretching tends to work much like linear progression or linear repetition progression works to allow you to progress in both strength and hypertrophy. If you think about it, stretching is like a progressive eccentric exercise. Rather than increasing strength, your goal is to normalize your body to an elongated muscle state when it wants to contract against you. When you allow your body to relax in the state where you are experiencing slight discomfort but not pain, you should progressively improve in your stretching.

It is normal that some people will not improve their stretching using standard static stretching, especially those with high nervous system sensitivity. Most athletes are stretching wrong, so definitely give it a try for a month or two before moving on to other methods. If permanent plateauing happens to you, know that you can change your programming methods for stretching in the same way you can change the programming for strength or hypertrophy training. Let's talk about some of the other techniques beyond simple linear progression that you can use for stretching.

PROPRIOCEPTIVE NEUROMUSCULAR FACILITATION

Proprioceptive neuromuscular facilitation (PNF) is another method that takes aim at the nervous system to relax the muscle. There are many techniques that fall under this umbrella, but the most common by far are the hold/relax and contract/relax methods. In these techniques, the athlete being stretched will contract or hold against another person or implement for a set period of time. After this, they lengthen their muscle and allow it to relax for a set period of time. This helps relax the muscle spindles at the edge of the range of motion, which may help people who are chronically tight.

The goal is to "reset" your muscles spindle sensitivity by application of force and then relaxation at your end range of motion. This will signal to your body that it is safe when you apply end range force, and it utilizes the body's natural reflex to relax muscles after a sustained contraction.

A typical PNF contract-relax cycle looks like this:

- Move the muscle to end range.
- Contract the muscle against light force for about ten seconds. The contraction should be isometric: the light force should equal the amount of force from the muscle so it does not move.
- After letting go of contraction, allow the muscle to relax. The light force should move your limb into a greater amount of stretching for about twenty seconds. Move into discomfort only and not pain.
- Repeat these steps for 2-5 different cycles.

Typically, 2-5 sets of 2-5 cycles of hold/relax or contract/relax are effective. The hold and contract phases can be anywhere from 5-15 seconds, depending on how tight your muscle is. 5-10 seconds for this phase is optimal. Remember, you have to fatigue the muscle to allow it to start releasing from its shortened length. The relax phase can be anywhere from 10-30 seconds, though 10-15 seconds for this phase is ample. Your total stretching time ends up being approximately 80 seconds with a contract phase of 10 seconds and a relax phase of 15 seconds.

Since breathing is intimately coupled with the nervous system and relaxation, deep breathing can also be used during the relaxation phase to allow your muscles to stretch. Remember that pain increases sensitivity in your muscle spindles, so pushing the stretch into painful territory is counterproductive. Instead, it is better to move to the end of the range where discomfort begins, contract the muscle, and relax into the range of the same amount of discomfort. It is inadvisable to start in the range of discomfort, contract your muscles, and then relax into a range where there is potentially more discomfort or pain.

If your muscles are extremely tight, an effective variation is to contract the opposing muscles during the relaxation phase. If you are stretching your hamstrings, contract your hip flexors and quadriceps in the relaxation phase. This is the phase where your hamstrings are lengthening. It is based on the principle of reciprocal inhibition, in which the body will naturally relax the muscles on the opposite side of the joint(s) as one muscle is activated.

A different method that may help with very tight muscles involves utilizing reflex arcs (such as the patellar reflexive arc) to inhibit your hamstrings. Tap your patellar tendon in the same way a doctor would to check your reflexes. Hitting it multiple times will allow the reflexive arc to activate and inhibit your hamstrings. Then you can perform a PNF-type stretch.

LOADED STRETCHING

While the concept of *loaded stretching* has existed for some time in various forms, this concept has gained greater traction in the past few years as a methodology for stretching. The first edition of *Overcoming Gravity* used a different terminology for this concept.

Science has known for some time that movements taken to the edge of their range of motion with weights are as effective or potentially more effective than static stretching. For example, if you are performing weighted squats you will want to descend to the bottom of the squat until you feel a stretch on the hamstrings. If you continue to perform squats this way you will naturally increase your range of motion, especially as the weights get heavier.

One way to implement this is known as the “Asian squat.” This involves going into the bottom of the squat position and remaining in that position while holding a weight. This can help you achieve a “deeper” position that ultimately improves flexibility. Along those same lines, preferentially shifting your weight around on your feet can stretch out your calves, hamstrings, glutes, and quadriceps while in the bottom of the squat. This will also increase your flexibility. This method essentially “loads” different muscles at the bottom of the squat to increase flexibility. If you are extremely tight, it can be more effective than PNF-type stretching to remain in the bottom of the squat position. Do it for as long as you can: five minutes or more.

In terms of progression, you can think of loaded stretching along the same lines as simple progression techniques. For example, this is how you may progress with this method of stretching:

- Perform one set of ten toe touches with a ten-second hold at the bottom to work on increasing hamstring and back flexibility.
- Once you stop progressing from session to session, hold two pounds of weights to allow you to be pulled deeper into the stretch. If necessary, stand on a raised surface to allow the hands to drop below the ground.
- After you stop progressing, consider adding more sets, weight, or time spent in the hold.
- Eventually, remove sets, weight, and time in order to obtain the flexibility gains with less loads.

The goal is not to reach a high amount of sets, weight, or time, but to move into a new range of motion and allow it to solidify. This may mean after you have increased your range of motion that you move down in sets, weights, or time spent in the stretch in order to maintain it. This would be analogous to deloading with exercise and beginning another cycle. The repetitions, sets, and load do not *have* to keep increasing to reach a new range of motion. Moving into a new range of motion without increasing these things tremendously is a sign that your overall flexibility has improved.

The only way to move into a new range of motion is to move to the edge of your current range and allow your body to conform to the length increases. You can think of this in terms of hypertrophy and time under tension. The more time you spend in a muscle-lengthened state without pain, the more your body will conform to accommodate that state. Here are the two ways to load stretches to increase flexibility:

- Isometrics at the edge of your range of motion
- Eccentrics through the range of motion to the end range

Isometric holds can be held at end range for sustained lengths of time from 30-60 seconds, increasing the repetitions of end range holds, loaded with incremental one-pound weights, or any combination of the above. Eccentrics can be performed with a uniform lowering through the movement (with or without a hold at the end), with increased repetitions, or loaded incrementally. Each of these will work; the effectiveness depends on how your body responds. Every person is different, so feel free to experiment and figure out what works best for you.

Even beyond PNF or loaded stretching, any type of system where you are practicing a lot of end-range movement will be effective for increasing your range of motion. Yoga is a prime example of this. The goal is working on the poses, not necessarily becoming more flexible, but it just so happens that moving into the poses requires many end-range muscle positions that will help you get more flexible. Active isolated stretching is another example.

When you understand the concept of end-range work, discomfort without pain, and nervous system relaxation, you understand how various systems allow you to make improvements in flexibility. This is why it is so important to understand the physiological processes behind how everything works. When you can troubleshoot your own workouts and flexibility work, you can discover the method that works best for you. You do not have to buy tons of programs, or go to a “guru” to help you make progress. You can learn to rely on your burgeoning experience with your own body.

Stretching too much is akin to programming too many exercises or putting too much volume in a workout. You become sore and may potentially lose any gains you have made. More is not always better. Focus on making small, incremental amounts of progress and maintaining the gains you have made.

Generally speaking, these are the concepts that help improve your overall flexibility:

- Time spent at end range (more time spent = more accommodation).
- Relax your nervous system through deep breathing, contract-relax, etc.
- Avoid pain by stretching only into the range of discomfort (for sensitive nervous systems).
- Increase strength through your range of motion and end range, especially by using a slow, controlled eccentric phase (unloaded or loaded stretching).

MAINTAINING FLEXIBILITY GAINS WITH MOBILITY AND FLEXIBILITY WORK

One of the concepts often left unaddressed is how to maintain a new range of motion once you have obtained it. Generally, you should expect increases in range of motion if you are foam rolling, massaging, or performing joint mobilization or flexibility work. This, of course, does not always happen. Sometimes, particular muscles or joints tighten back up. While the previously mentioned issues of instability, weakness, or pain may inhibit one from decreasing tight musculature, there could be another culprit.

Flexibility and mobility are similar, but not interchangeable. Flexibility is about increasing range of motion or muscle length by means of stretching. It is difficult to move a joint when the muscles surrounding that joint are not able to stretch. If you cannot touch your toes, performing flexibility work will elongate your hamstring muscles until you can.

Mobility is an umbrella term that covers any movement within an existing range of motion. Typically, it is performed at lower intensities with just your bodyweight. Mobility has three specific goals: maintaining existing ranges of motion, improving motor learning, and increasing movement quality. If movement within an existing range of motion is loaded with weight or resistance at a high enough one repetition maximum, it will become strength or hypertrophy work.

Whenever you obtain a new range of motion, you should follow up by performing both passive and active mobility work in the new range. For example, if you stretched your ankle, you will want to follow it up with passive mobility work that consists of moving your ankle in the new range manually, then actively contracting your muscles in the new range. This helps to solidify your gains by providing feedback to the body that the new range of motion is safe and can be used effectively. Once this happens, your body will no longer send mechanoreceptor (pressure/distortion) or nociceptor (pain) feedback to your nervous system that causes your muscle spindles to sensitize and limit range of motion.

Habits are hard to break. If your body has not been flexible for years, it tends to revert to what it knows. Your muscle spindles may start to naturally sensitize even if you have not stretched into pain. This can happen very easily if you do not perform mobility or range of motion work. This is the same as changing your posture. If you have had poor alignment for years, moving into a different alignment is going to be uncomfortable for a few weeks as your body adjusts to the new position. If you are seeking to make your body flexible for the first time in your life, you must get in the habit of stretching and performing mobility work on a daily basis, or

even multiple times a day. This may be difficult at first, but once the new habit is established, you will be able to maintain your gains much more easily.

If you do nothing to maintain your range of motion, your body will tighten back up. This is why mobility and flexibility work outside of a routine is so important. Most people who are flexible and mobile will be training these *every* day, not just on workout days. If you perform mobility work daily and flexibility work every few days, you should be able to maintain your range of motion gains. However, if you want to progress to the next range of motion quicker, performing both mobility and flexibility work on a daily basis is recommended. The phrase *Use it or lose it!* results from the SAID principle. If you do not use your new range of motion, you will likely lose it before your next workout.

It is important to construct your workout routine with *all* your goals in mind, not just the workout itself. A routine designed to increase flexibility may only have you stretching a couple times a week; however, you really need to be performing mobility *and* flexibility work daily to reach your goal of becoming more flexible. This may seem unmanageable, but there are many ways to make mobility and flexibility a part of your daily routine, even if you keep a busy schedule. You can perform a little when you wake up and before you go to bed at night, or as you are preparing a meal. Whether or not you perform both mobility and flexibility daily depends on your goals, but the bottom line is you need to do something every day.

Some people will be able to progress in flexibility while only stretching three times per week. Others may need to stretch multiple times per day, seven days per week. Then there are those who may need to stretch three to five times per week in addition to performing mobility work multiple times per day. If you notice your routine is not working, vary it to figure out what works best for you! Additionally, you can begin using PNF or LS to plan your flexibility workouts.

COMMON AREAS THAT NEED PREHABILITATION AND FLEXIBILITY WORK

- If your joints and connective tissues start to feel stressed, I would dial down on high-difficulty exercises and begin moving toward a greater volume (such as 12-20 repetitions) near the beginning of your routine as a warm-up. Add general ankle and foot mobility and stretching as needed for other sports or strength.
- For hip and leg flexibility, you will want to work on splits, straddle, and pike position with the ability to get your stomach to the floor.
- For your back, you will want to master bridges and mobilization of the thoracic spine if it is tight.
- For your shoulders, you will want to work on lats, chest, and scapular flexibility and mobility so that you can get your arms fully overhead and into hyperextension.
- For your elbows, you simply need a reasonable range of motion: fully straight, 180 degrees. You may need RTO supports or specific biceps work if they hyperextend.
- For your wrists, you want to be mobile in every direction because you will use them for almost everything.

Specific joints have no ideal number of degrees that mobility or flexibility should reach. One person is not as flexible as the next. However, you should be able to move into any of the typical stretching exercises like splits, German hangs, wrist pushups, overhead straight-arms for handstands.

In an exercise such as the German hang (stretching your shoulders far behind you), you want to be able to stay comfortably in that position for at least 20-30 seconds. It doesn't matter if this is at a 160-degree angle beyond neutral or 120-angle; both are fine. What matters is that your range of motion allows you to move in and out of the position comfortably. This means you are both mobile and stable; therefore, you can effectively develop strength.

If your joints are getting sore, you have trouble moving through the full range of motion, or you have difficulty applying strength near the end of the range of motion, you will definitely want to use prehabilitation work to correct the issue. This should be performed at the end of your routine, when your tissues are adequately warmed up, so the stretching and mobility work will be very productive. Do not ignore this, as it is important to stay healthy and this is one of the long-term factors of making progress. You cannot expect to become strong if you continually beat up your body through rigorous exercise without caring for it as well.

CHAPTER 11 SUMMARY

PREHABILITATION, ISOLATION, FLEXIBILITY, AND COOL DOWN

Prehabilitation refers to any work that is focused on injury prevention. Prehabilitation exercises are focused on correcting existing imbalances or preventing injuries from forming with specific exercises that address various weak links. Rehabilitation work can be done at this point in a workout, but it is also viable in separate, stand-alone sessions.

Isolation work is used for multiple purposes: as prehabilitation work to strengthen connective tissues, specifically to strengthen or hypertrophy muscles for aesthetics, or it can be used in combination with prehabilitation and/or strength work to simultaneously strengthen your body and prevent injury by correcting imbalances and bringing up specific weak links. Prehabilitation and isolation work will usually have some overlap and can be used for the same purposes.

Flexibility work and *cool down* are the last part of the routine. While flexibility work is recommended, the cool down can have additional light activity such as jogging, jumping rope, mobility work, or other types of movement. Additionally, if you have an oddball exercise that does not naturally fit into a routine, you can place it here.

There are three different methods to improve flexibility:

- Regular Stretching—standard static stretching; 3 sets of 30-60s holds.
- Proprioceptive Neuromuscular Facilitation—usually contract/relax, hold/relax, or contract/pull. Improves range by relaxing the muscle after sustained contraction and reciprocal inhibition.
- Loaded Stretching—used to progressively pull the body further into a stretch.

Any of these three options should be performed while breathing deeply to relax your body. Ensure that you do not cross from the range of discomfort to the range of pain. Pain sensitizes muscle spindles, which reduces flexibility in the long run. Measure flexibility effectiveness using *time spent at end range*, which is much like *time under tension*, a similar hypertrophy method.

Finally, if you want to maintain and improve the gains you have made within a range of motion, *use it or lose it!* You should be doing mobility and flexibility work every day—multiple times a day if you are looking to break into a new range. The frequency you select should reflect your individual goals.